

Rival Moh. Wahyudi

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SUMMARY

I'm an Information Systems student in Universitas Negeri Semarang who is passionate about AI/ML, data science, and analytics. I enjoy working across the full data pipeline: exploring messy real-world datasets, creating visualisations to uncover patterns, and building models that solve meaningful problems. My interests span time series forecasting, data analysis, computer vision, and hybrid modelling. I'm especially excited about using data science to support public sector innovation and sustainable solutions.

EDUCATION

Universitas Negeri Semarang

Aug 2023 – Present

Undergraduate Information System Student

GPA: 3.67/4.00

Expected graduation June 2027

PROJECT

Student Dropout Predictive Pipeline & Early Warning System | Personal Portfolio Project March 2026
Tech Stack & Methodologies: Python (Pandas, Scikit-Learn, XGBoost), SQL (SQLite), ETL Pipeline Architecture, Relational Database Design (3NF), Explainable AI (SHAP), Feature Engineering.

- Architected an end-to-end ETL pipeline in Python to ingest, clean, and normalize flat educational datasets into a structured 3NF SQLite database, ensuring strict relational data integrity for downstream machine learning.
- Diagnosed and purged severe data leakage during baseline testing, sacrificing raw overall accuracy to build a mathematically honest “Early Warning System” constrained entirely to Semester 1 data, securing a 6-month intervention window for administrators.
- Trained and optimized an XGBoost classification engine focused strictly on minority class recall rather than basic accuracy, successfully capturing 87% of at-risk students before their second semester began.
- Integrated Explainable AI (SHAP) to translate black-box model predictions into transparent, student-specific waterfall plots, enabling academic counselors to allocate retention resources based on proven, localized mathematical drivers.

SuaraWarga Voice-First Disaster Reports | IDCamp 2025: Small Apps for Big Preparedness Feb 2026

- Engineered a voice-first Progressive Web App (PWA) using Python (FastAPI) and JavaScript, enabling citizens to report hands-free disasters in high-stress, low-bandwidth environments.
- Integrated Google Gemini 1.5 Flash (Multimodal LLM) to automatically transcribe noisy audio and extract structured data (hazard type, severity, location) with custom prompt guardrails to prevent AI hallucinations.
- Developed an audio pre-processing and geospatial pipeline utilizing librosa and noisereduce to clean background noise, paired with geopy (Nominatim) and HTML5 Geolocation to map incidents dynamically via Leaflet.js.
- Architected resilient, lightweight backend logic that handles asynchronous audio uploads, reverse geocoding for LLM context grounding, and persistent data storage using SQLite.

WORK EXPERIENCE

Wiyora Technology - Semarang

Oct 2025 – Present

Intern – Machine Learning & Data Pipeline Development (Ongoing)

- Building the automated data-collection pipeline using Python and Playwright to crawl Twitter, Facebook, and Instagram on a scheduled basis.
- Preparing and cleaning collected data for downstream sentiment-analysis model development in an ongoing multi-module web project.

Outlier – San Francisco

Jan – Mar 2025

Freelance AI-Language Trainer (Indonesian Writer)

- Evaluated and ranked over 10 AI-generated responses to ensure accuracy and consistency, while providing constructive feedback and refining training prompts to enhance the AI's natural language

capabilities according to client specifications.

- Engaged in remote collaboration for data annotation and analysis of AI outputs, focusing on evaluating their accuracy, relevance, and adherence to stylistic guidelines, with the goal of reducing instances of model "hallucinations."

ADDITIONAL EXPERIENCE

I-Secret of Computer Science – Semarang, Indonesia

Mar 2024 – Feb 2025

Larut Eksternal

- Invited and coordinated over 40 students to enhance the learning community in emerging technologies. Collaborated with peers to design workshop materials and manage logistics for a smooth educational experience.

PT Petrokima Gresik – Gresik, Indonesia

Sep 2021 – Aug 2022

Internship Management Staff in the production Partner Management Division

- Coordinated with 150+ partner mitras to meet production goals and quality standards. Assisted in analysing production data and evaluating performance for process improvements.

SKILLS

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- **Technical Skills:** Python, SQL, Machine Learning, Pandas, NumPy, TensorFlow, Scikit-Learn, Matplotlib, AI, Data Analysis, Data Science, Data Visualization, Data Processing, PostgreSQL, ETL/ELT, Streamlit, Looker Studio, Tableau.
 - **Interpersonal Skills:** Time Management, Problem Solving, Adaptability, Analytical Thinking, Leadership, Communication, Teamwork, Attention to detail, Organization.
 - **Language:** Indonesian, English.

CERTIFICATE

Dicoding Machine Learning Engineer

2024 – 2025

- **Certification of Completion of the Practical Machine Learning Class** Jan 2025
Skill Acquired: ML System Design, Project Report, Predictive Analysis, Sentiment Analysis, Computer Vision, Recommendation System.
- **Certification of Completion of Data Analysis with Python Class** Aug 2024
Skill Acquired: Data Analysis, Data Processing, Data Wrangling, Exploratory Data Analysis (EDA), Data Visualisation, Dashboard (Streamlight)
- **Certification of Completion Machine Learning Development Class** Aug 2024
Skill Acquired: Deep Learning, Neural Network, NLP, Time Series, Image Classification, Reinforcement Learning, ML Convert.
- **Certification of Completion of the Machine Learning Basic Class** Jul 2024
Skill Acquired: Data Analysis, Data Processing, Data Wrangling, Exploratory Data Analysis (EDA), Data Visualisation, Dashboard (streamlit)
- **Certification of Completion SQL Basic Class** Jun 2024
Skill Acquired: Data, DBMS, Data Manipulation Language, Basic Query